CODING5S ECOSYSTEM | PEDAGOGY QUICK READ

Formula Auto-Injector — Prompt Formulas Paster

Batch injection engine for the Coding5s Creator Kit pipeline: automated extraction, validation and multi-range deployment of prompt-generation formulas across five modular stages.

EXECUTIVE SUMMARY

The **Formula Auto-Injector** is a Windows-based COM automation engine built for the Coding5s educational framework. It replaces a manual, error-prone workflow — copying generator ranges, cleaning formatting in Notepad, and manually extending formulas to row 200 — with a single-click, atomic batch operation performed directly on the **PromptGenerator** master sheet of a target Creator Kit.

Operating through an isolated, invisible Excel instance (`DispatchEx`), the engine reads five modular generation sheets (`FGen_S1`–`FGen_S5`), assembles formula strings in memory, and injects them into full destination ranges in one native operation — letting Excel's own engine auto-scale relative references across all 200 rows. A hard-coded **8,100-character safety guard** protects the workbook against Excel's 8,192-character formula limit, aborting all changes instantly if any stage would exceed it.

5	200	8,100	1-Click
GENERATION STAGES	ROWS AUTO-SCALED	CHAR SAFETY LIMIT	EXECUTION

SYSTEM ARCHITECTURE & COMPONENTS

```
pillar_1/coding5s_creator_kits/
│
├── copy_paster_Prompt_Formulas_for_CK.py  Core injector engine
├── Creator Kit Updater Runner.bat       1-click execution script
├── formula_injector_guide.md           Documentation
└── Coding5s ... Creator Kit.xlsx         Target workbook
```

Two-component pipeline:

- **Python Engine** — handles independent COM instantiation, coordinate reading, character validation and batch injection into `PromptGenerator`.
- **Batch Runner** — triggers a native VBScript confirmation dialog describing the macro's scope before launching the Python engine.

AUTO-INJECTION PIPELINE — SEQUENTIAL STAGES

STAGE 01 — File Existence Validation

OBJECTIVE

Confirm the target Creator Kit exists in the local working directory before any operation begins.

KEY BEHAVIOR

- Reads `ARCHIVO_CREATOR_KIT` from the configuration block.
- Prints a critical error and exits safely if the file is missing — no partial execution.

STAGE 02 — Isolated Background Engine Initialization

OBJECTIVE

Launch a fully independent Excel process in memory without disturbing the user's active session.

KEY BEHAVIOR

- Uses `DispatchEx` — **Visible = False**, **ScreenUpdating = False**, **Interactive = False**.
- Eliminates interface flicker and window-stealing during batch runs.

STAGE 03 — Coordinate Mapping & Formula Assembly

OBJECTIVE

Rebuild each formula string exactly as a manual Notepad clean-up pass would, but entirely in memory.

KEY BEHAVIOR

- Iterates `MAPA_FORMULAS` across Topics, Projects and Mentors for all 5 stages.
- Reads source ranges (e.g. `E3:E50`), discards empty cells, joins with newline characters.

STAGE 04 — Safety Verification & Range Injection

OBJECTIVE

Guarantee zero workbook corruption while writing formulas natively across full destination ranges.

KEY BEHAVIOR

- Halts instantly and closes without saving (`wb.Close(False)`) if length > 8,100 chars.
- Batch-assigns the compiled string to ranges like `K9:K200` in one atomic write.

STAGE 05 — **Native Save & Memory Cleanup**

OBJECTIVE

Persist all changes and release system resources cleanly once every stage passes validation.

KEY BEHAVIOR

- Saves via `wb.Close(True)`; restores screen updating and application events.
- Quits the background Excel process, leaving no orphaned tasks in memory.

TARGET CONFIGURATION

```
# ==========================================
#    MAIN CONFIGURATION
# ==========================================
ARCHIVO_CREATOR_KIT = "Coding5s Python Core & Scripting v0.4 Creator Kit.xlsx"
LIMITE_CARACTERES = 8100
PESTANA_DESTINO = "PromptGenerator"
FILA_FIN = 200
```

PARAMETER	PURPOSE
`ARCHIVO_CREATOR_KIT`	Exact filename of the Creator Kit being edited
`LIMITE_CARACTERES`	Safety ceiling below Excel's 8,192-char hard limit
`PESTANA_DESTINO`	Destination sheet — **PromptGenerator**
`FILA_FIN`	Final row for range auto-scaling (row 200)

SETUP & REQUIREMENTS

System Requirements

- Windows 10 / 11
- Microsoft Excel installed locally
- Python 3.8+

Dependencies

```
pip install pywin32
```

Provides the Windows COM extension package required for `DispatchEx`.

STEP-BY-STEP EXECUTION

STEP	ACTION
1. Configure	Set `ARCHIVO_CREATOR_KIT` to the exact target filename and save the script.
2. Verify Placement	Ensure the script, batch runner and Creator Kit reside in the same directory.
3. Execute	Double-click `Creator Kit Updater Runner.bat` and confirm the native prompt.
4. Monitor	Console displays real-time injection metrics and character counts per stage.
5. Verify Output	Open the Creator Kit to confirm all prompts across all stages render correctly.

TECHNICAL FOOTER

Architectural reference for engineers integrating or maintaining the Formula Auto-Injector.

KEY ARCHITECTURAL PILLARS

- **Isolated COM Automation** — independent `DispatchEx` Excel instance, no UI interference.
- **Native Dynamic Scaling** — full-range targeting instead of simulated drag, using Excel's own reference engine.
- **Atomic Batch Injection** — one write operation per destination range.

SYSTEM GUARANTEES

- **8,100-char guard** below Excel's 8,192 hard limit, checked before any write.
- **Fail-safe abort** — `wb.Close(False)` discards changes on violation.
- **Clean memory teardown** — screen updating, events and process fully restored.

REFERENCES

- `copy_paster_Prompt_Formulas_for_CK.py` — core engine
- `Creator Kit Updater Runner.bat` — execution trigger
- `formula_injector_guide.md` — source documentation
- Dependency: `pywin32` (COM bindings)

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